



Mean–variance approximations to expected utility[☆]

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ABSTRACT

It is often asserted that the application of mean–variance analysis assumes normal (Gaussian) return distributions or quadratic utility functions. This common mistake confuses sufficient versus necessary conditions for the applicability of modern portfolio theory. If one believes (as does the author) that choice should be guided by the expected utility maxim, then the necessary and sufficient condition for the practical use of mean–variance analysis is that a careful choice from a mean–variance efficient frontier will approximately maximize expected utility for a wide variety of concave (risk-averse) utility functions. This paper reviews a half-century of research on mean–variance approximations to expected utility. The many studies in this field have been generally supportive of mean–variance analysis, subject to certain (initially unanticipated) caveats.

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I wish to thank the editors of this special edition of the European Journal of Operations Research, commemorating the 60th anniversary of Markowitz (1952), for inviting my comments on this occasion. For the most part the past sixty years have been a source of great satisfaction for me. The idea I expressed in 1952, and more completely in 1959, has been filled out by countless insightful and energetic people concerned with, for example, the estimation of means, variances and covariances including the use of factor and scenario models for such purposes; top down versus bottom up processes for use by practitioners; data to serve as grist for various estimation mills; the treatment of estimation uncertainty; the consequences of everyone following the prescriptions of Markowitz (1952) or Markowitz (1959); etc. My first-draft attempt to acknowledge a few major contributors in each of these areas convinced me that I was bound to leave out at least dozens of people who should be acknowledged, and I should therefore not attempt such a task. I will instead simply issue a general—but heart-felt—“Thank You” to all those who have helped transform a thought into a field of great practical importance.

However, there is one area where many brilliant scholars give bad advice because they are confused as to the conditions under which mean–variance analysis is applicable. I refer to this as our field’s Great Confusion—namely the confusion between necessary and sufficient conditions for the practical use of mean–variance

analysis. For example, Gaussian return distributions¹ or quadratic utility are sufficient conditions for the use of mean–variance analysis in practice. But they are not necessary conditions. In particular, if you believe that rational decision making may be characterized by axioms which imply that one should maximize expected utility using probability beliefs where objective probabilities are not known (as Savage (1954) persuaded many including me), then the necessary and sufficient condition for the practical use of mean–variance analysis is that a careful choice from a mean–variance efficient frontier will approximately maximize expected utility for a wide variety of concave (risk-averse) utility functions.

For example, Markowitz (1959) observes that if a utility function, $U(R)$, can be approximated closely enough by a quadratic for a sufficiently wide range of returns, then EU will be approximately equal to some function, $f(E, V)$, of expected return E and variance of return V . In particular, Markowitz (1959) suggests two methods for approximating a utility function by a quadratic, namely:

$$Q_Z(R) = U(0) + U'(0)R + 0.5U''(0)R^2 \quad (1)$$

$$Q_E(R) = U(E) + U'(E)(R - E) + 0.5U''(E)(R - E)^2 \quad (2)$$

For the natural logarithm utility function, $U = \ln(1 + R)$, approximations (1) and (2) are, respectively,

$$q_Z(R) = R - \frac{1}{2}R^2 \quad (3)$$

$$q_E(R) = \ln(1 + E) + (R - E)/(1 + E) - (R - E)^2/[2(1 + E)^2] \quad (4)$$

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¹ More generally, if a probability distribution is joint elliptical then EU is a function of mean and variance. See Chamberlain (1983).

Table 1
Comparison of $\ln(1+R)$ with $R-1/2R^2$.

R	$\ln(1+R)$	$R-1/2R^2$
-0.50	-0.69	-0.63
-0.40	-0.51	-0.48
-0.30	-0.36	-0.35
-0.20	-0.22	-0.22
-0.10	-0.11	-0.11
0.00	0.00	0.00
0.10	0.10	0.10
0.20	0.18	0.18
0.30	0.26	0.26
0.40	0.34	0.32
0.50	0.41	0.38

The expected values of Eqs. (3) and (4) are functions of mean and variance, namely,

$$f_Z(E, V) = E - (E^2 + V)/2 \quad (5)$$

$$f_E(E, V) = \ln(1+E) - V/[2(1+E)^2] \quad (6)$$

As shown in Table 1 here, which is Table 2 on Page 121 of Markowitz (1959), for returns R between a 30% loss and a 40% gain on the portfolio-as-a-whole there is little difference between $\ln(1+R)$ and q_Z . At $R = -0.30$ (a 30% loss) $\ln(1+R) = -0.36$ whereas the quadratic is -0.35 . At $R = 0.40$ (a 40% gain) $\ln(1+R) = 0.34$ whereas the quadratic is 0.32 . Between these two values, i.e., for $R = -0.20, -0.10, \dots, +0.30$, the approximation equals the log utility function to the two-places shown. Even at a 40% percent loss or a fifty percent gain, the difference is noticeable but not great: -0.51 vs. -0.48 in the one case; 0.41 vs. 0.38 in the other. Thus Markowitz (1959) concludes that for choice among return distributions which are mostly within the range of a 30% loss to a 40% gain on the portfolio-as-a-whole, and do not fall outside this range “too far, too often,” the $E[\ln(1+R)]$ maximizer will almost maximize expected utility by an appropriate choice from the mean-variance efficient frontier.

Subsequent research, reported below, shows that f_E is a much better approximation to $E\ln(1+R)$ than is f_Z , and that f_E approximates fairly well for historical return distributions which are much more spread out than Table 1 would suggest. This is because q_E errs in one direction if $R-E < 0$ and in the other for $R-E > 0$. These errors tend to cancel.

Various authors have tested the ability of functions of mean and variance to approximate expected utility for various utility functions and historical or postulated portfolio return distributions. By and large their conclusions have been favorable. But most

Table 2
Correlations between EU and $f_{0.01}(E, V)$ for four historical distributions, 1958–1967.

Utility function	Annual returns of 149 mutual funds	Annual returns on 97 stocks	Monthly returns on 97 stocks	Random portfolios of 5 or 6 stocks
$\log(1+R)$	0.997	0.880	0.995	0.998
$(1+R)^a$				
$a = 0.1$	0.998	0.895	0.996	0.998
$a = 0.3$	0.999	0.932	0.998	0.999
$a = 0.5$	0.999	0.968	0.999	0.999
$a = 0.7$	0.999	0.991	0.999	0.999
$a = 0.9$	0.999	0.999	0.999	0.999
$-e^{-b(1+R)}$				
$b = 0.1$	0.999	0.999	0.999	0.999
$b = 0.5$	0.999	0.961	0.999	0.999
$b = 1.0$	0.997	0.850	0.997	0.998
$b = 3.0$	0.949	0.850	0.976	0.958
$b = 5.0$	0.855	0.863	0.961	0.919
$b = 10$	0.447	0.659	0.899	0.768

authors—including almost all textbook writers—seem ignorant of this body of literature. In particular, the fact that Markowitz (1959) bases its support of mean-variance analysis on mean-variance approximations to expected utility is rarely cited. For example, Quiggin (1993) says, “The EU approach initially faced strong competition from mean-variance analysis, exemplified by the work of Markowitz (1959) on portfolio analysis, but the logical foundations of this approach were far more dubious than those of expected utility theory.” [p. 8]. Examination of the *Table of Contents* (!) of Markowitz (1959) would have revealed a chapter titled “Expected Utility.” A cursory examination of this chapter would have shown that the premises of utility analysis and the premises which Markowitz (1959) proposed in support of mean-variance analysis were identical to those of EU.

Quiggin is not alone in this mistaken notion about the fundamental assumptions of portfolio theory. This error is widely held despite my repeated attempts to spread the truth on this subject (as in Levy and Markowitz, 1979; Kroll et al., 1984; Markowitz, 1991, 1997, 2010, 2012a, forthcoming; Markowitz et al., 1994; Markowitz and Usmen, 1996a,b, 2003; Markowitz and van Dijk, 2003), and despite various post-Markowitz (1959) research into the efficacy of mean-variance approximations to expected utility, most of which is supportive of the Markowitz (1959) view.

The remainder of this paper surveys such research, with attention to negative as well as positive reports.

1. Why not just maximize expected utility?

If one believes that action should be in accord with the max EU rule, why seek to *approximately* maximize it via a mean-variance analysis? Why not just maximize expected utility? In addressing this question I distinguish three types of expected utility maximization:

- explicit,
- MV-approximate,
- implicit.

I refer to it as “*explicit*” EU maximization when a utility function is given and analytical or numerical methods are used to find the portfolio that maximizes the expected value of this function. In contrast, I refer to it as “*MV-approximate*” when a utility function is given and a mean-variance approximation to its expected utility is maximized. An example would be to approximately maximize $E\ln(1+R)$ by generating an MV efficient frontier and choosing from it the portfolio that maximizes f_Z or f_E in Eq. (5) or (6).

As reviewed below, Levy and Markowitz (1979) finds that mean-variance approximations are usually quite accurate. From this it concludes, for some hypothetical investor Mr. X, that “If Mr. X can carefully pick the MV efficient portfolio which is best for him, then Mr. X, who still does not know his current utility function, has nevertheless selected a portfolio with maximum or almost maximum expected utility.” I refer here to such a process as “*implicit*” expected utility maximization.

Typically it is much more convenient and economical to determine a set of mean-variance efficient portfolios than it is to find the portfolio which maximizes expected utility. Historically, one source of inconvenience and added expense for the latter was computational. One typically had to wait longer (perhaps hours longer) and pay a higher computer bill to find an expected-utility-maximizing portfolio than to trace out a mean-variance frontier. This computational problem is now trivial thanks to faster, cheaper computers. It still takes many times as long to compute the expected value of most concave functions as it does to trace out a

mean–variance efficient frontier.² But neither calculation takes long enough to be a practical limitation.

There are, however, other expenses and inconveniences that remain for explicitly maximizing expected utility as compared to using an MV or implicit approximation to it. The first of these concerns parameter estimation. The only inputs required for a mean–variance analysis are the means, variances and covariances of the securities or asset classes of the analysis. (A factor model can serve in place of individual variances and covariances.) This is because the formulas relating the expected return and variance of a portfolio to the expected values, variances and covariances of return of securities do not depend on the form of the probability distribution. Letting E_p be the expected return of the portfolio; X_i , the fraction of the portfolio invested in the i th security; and E_i the expected return of the i th security or asset class, the relationship

$$E_p = \sum_{i=1}^n X_i E_i \quad (7)$$

is true whether or not returns are normally distributed, or even symmetrical and whether or not the return distributions have “fat tails,” as long as the E_i exist and are finite. Similarly, letting V_p be the variance of the portfolio, and σ_{ij} be the covariance between security returns r_i and r_j , the formula for portfolio variance

$$V_p = \sum_{i=1}^n \sum_{j=1}^n X_i X_j \sigma_{ij} \quad (8)$$

is true whether or not the return distributions are symmetrical or have fat tails, as long as the V_i are finite.

The case is different when one explicitly maximizes expected utility. Then one must determine what type of joint probability distribution generates return combinations, $(r_1, r_2, \dots, r_n)$, and must estimate the parameters for such a joint distribution. Accomplishing this can be a substantial research project.

A second difficulty with using explicit expected utility maximization, as opposed to implicit EU maximization, is that someone must determine the investor's utility function. As Von Neumann and Morgenstern (1944) explain, theoretically this should be done by having the investor answer a series of questions as to what probabilities p_a of returns R_a versus $(1 - p_a)$ of R_c the investor considers just as desirable as return R_b with certainty, for various returns R . This would be challenging enough for an institutional investor, such as an endowment or pension fund with a single large portfolio, but seems hardly possible in any thorough way on behalf of the many clients of a financial advisor. This step is not necessary when implicit EU maximization is used.

Finally, another advantage of using implicit EU maximization is that no one has to explain the expected utility concept to the individual investor, or to the supervisory board of an institutional investor, or to the typical financial advisor. Instead, portfolio choice can be couched in the familiar terms of risk versus return.

2. Utility of return vs. utility of wealth

Eqs. (3) and (4) assume that an investor seeks to maximize the expected value of $\ln(1 + R)$, where R is the return on the portfolio

during the forthcoming period.³ Instead suppose one assumed that the investor seeks to maximize the expected value of some function of end-of-period wealth W_{t+1} . The latter is connected to return and beginning of period wealth W_t by the formula

$$W_{t+1} = (1 + R)W_t \quad (9)$$

In the case of a log utility function, the portfolio that maximizes $ELn(W_{t+1})$ also maximizes $ELn(1 + R)$, since Eq. (9) implies

$$Ln(W_{t+1}) = Ln(1 + R) + Ln(W_t) \quad (10)$$

When the portfolio to be held from time t to $t + 1$ is chosen, W_t is already given, therefore $Ln(W_t)$ is a constant as far as this choice is concerned. As von Neumann and Morgenstern explained, if two utility functions, U and V , are related linearly

$$V = a + bU \quad b > 0 \quad (11)$$

the probability distribution which maximizes EU also maximizes EV . Thus Eq. (10) implies that the same portfolio is chosen among those available whether one maximizes $Ln(1 + R)$ or $Ln(W_{t+1})$.

Next, consider the power utility function

$$U = X^a \quad \text{for } a > 0 \quad (12a)$$

or

$$U = -X^a \quad \text{for } a < 0 \quad (12b)$$

For example

$$U = X^{1/2} \quad (12c)$$

or

$$U = -X^{-1/2} \quad (12d)$$

As with the log utility function, a power utility function implies the same choices among probability distributions whether one let's X be W_{t+1} or $(1 + R)$ in Eqs. (12a) and (12b), since, e.g.,

$$(W_{t+1})^a = (1 + R)^a W_t^a \quad (13)$$

As above, W_t is already determined when the probability distribution of R (or W_{t+1}) is chosen, therefore the condition in Eq. (11) applies.

Eqs. (10) and (13) also imply that—for the power or log utility functions—the choice of portfolio does not depend on the level of start-of-period wealth W_t . This is not true for all utility functions. In particular it is not true of the exponential utility function

$$\begin{aligned} U &= -\exp\{aW_{t+1}\} \\ &= -\exp\{aW_t(1 + R)\} \end{aligned} \quad (14)$$

since this is of the form

$$U = -\exp\{-b(1 + R)\} \quad (15)$$

where b depends on W_t . A subsequent section will illustrate the dependence on b of choice among probability distributions by an investor with an exponential utility function.

If Eq. (14) were the investor's utility of wealth function, and remained constant through time, the investor's single-period utility of return function in Eq. (15) would change from one period to the next. Mean–variance approximation has no difficulty with this. The basic assumption is *not* that $U(R)$ stays the same with time, but that at any one time EU can be adequately approximated by some

² In addition to implying risk-averse behavior—preferring certainty to a gamble with equal or less expected return—the expected values of concave functions also have the property that a local maximum is also a global maximum, i.e., if some set of parameter values: $X_1 = X_1^0, X_2 = X_2^0, \dots, X_n = X_n^0$, have a greater value of EU than do any nearby values of $X_1, X_2, \dots, X_n$, then there is no distant set of values $X_1 = X_1^1, X_2 = X_2^1, X_n = X_n^1$, that gives a still greater value of EU. This greatly reduces computational time for maximizing EU.

³ In the financial literature, $(1 + R)$ is sometimes called “return” and R itself is called “rate of return”. Also, $(1 + R)$ is sometimes referred to as wealth, i.e., end-of-period wealth, W_{t+1} , scaled so that beginning-of-period wealth, W_t , is one. These various usages are equivalent, as long as used consistently in any one context. I will refer to R as either return or rate of return, interchangeably, since common English does speak of either a 6% return or a 6% rate of return. I refer to $(1 + R)$ as “one plus return”.

function of E and V . In particular, $U(R)$ could change with the investor's age and circumstances as well as wealth.

3. Loistl's erroneous analysis

In a paper titled “The Erroneous Approximation of Expected Utility by Means of a Taylor's Series Expansion: Analytical and Computation Results,” Loistl (1976) concludes that “the mean–variance approximation is not a good approximation of the expected value of utility at all.” [p. 909]. The reason Loistl reaches such a negative conclusion is as follows: note that in Table 1 a 6% return is represented as $R = 0.06$, not as $R = 6.0$. In particular, in Eq. (9), if starting wealth is $W_t = \$1,000,000$ and the portfolio returns 6%, then ending wealth is

$$W_{t+1} = 1,000,000 \cdot (1 + .06) = 1,060,000$$

NOT

$$W_{t+1} = 1,000,000 \cdot (1 + 6) = 7,000,000$$

Loistl assumes that utility U is a function of a random variable y “where y is the investor's wealth.” [p. 905]. He gives an example in which “Return in Percent” is listed as $-20, -10, \dots, 30, 40$. He then compares $U(y)$ with different Taylor series approximations, including the quadratic, fit with $X_0 = \bar{y}$, for $y = -20, y = -10, \dots, y = 40$, confining the comparisons to $y > 0$ for the logarithmic utility function. In other words, Loistl represents a 30% gain by $R = 30$ rather than $R = 0.3$. Loistl's conclusion—that the quadratic approximation fits poorly for these values of y —is consistent with the Markowitz (1959) observation that a quadratic fits well for returns in the interval -0.3 to 0.4 . Since a 30% gain must be represented by $R = 0.3$ rather than $R = 30$ in the calculation end-of-period wealth, W_{t+1} , as in Eq. (9), or its end-of-period expected utility as in Eq. (10), Loistl's negative conclusion about the mean–variance approximation is due to an erroneous analysis rather than an erroneous approximation.

4. Levy and Markowitz (1979)

The Levy–Markowitz study had two principal objectives:

- (1) to see how good mean–variance approximations are for various utility functions and portfolio return distributions; and
- (2) to test an alternate way of estimating expected utility from a distribution's mean and variance.

The Levy–Markowitz “alternate way” was to fit a quadratic approximation to U at three values of R :

$$(E - k\sigma), \quad (E), \quad (E + k\sigma)$$

They tried their approach for

$$k = 0.01, 0.1, 0.6, 1.0 \text{ and } 2.0$$

Of these, $k = 0.01$ did best in almost every case. This is essentially the same as the approximation in Eq. (2). I will therefore relate their results for $k = 0.01$ and subsequently treat these as if they were results for Eq. (2) approximation.

Table 2 here shows the Levy–Markowitz results for four data sets. The first column of the table lists various utility functions. The next shows results based on the annual returns for 149 mutual funds for the years 1958 through 1967—all the mutual funds whose returns were reported for the full period in the then latest edition of Wiesenberger (1941). Levy and Markowitz considered these 149 return series as 149 real-world return distributions. Specifically, the table's second column presents correlations between average utility

$$EU = \sum_{t=1}^T U(r_t) / T \quad (16)$$

and the mean–variance approximation $f(E, V)$ based on the quadratic fit through the three points with $k = 0.01$.

The utility functions used were the logarithm, and the power and exponential functions for the values of a and b shown in the table. For the logarithmic utility function, and for all the power utility functions considered, the correlation (over the 149 return distributions) between average utility and the mean–variance approximation to average utility was at least 0.997. Since this is more precision than one should expect from forward looking estimates of means, variances and covariances for a mean–variance analysis—or from estimates of joint distributions for an explicit expected utility maximization analysis—Levy and Markowitz conclude that, for these utility functions and return distributions, for all practical purposes EU and its mean–variance approximation are indistinguishable.

On the other hand, MV approximation was much less successful for exponential utility

$$U = -\exp\{-b(1 + R)\}$$

for $b = 5$ and, especially, for $b = 10$. We will consider such utility functions at length below.

The other columns of Table 2 show the correlation between EU and $f_{.01}$ for three more sets of historical distributions reported by Levy and Markowitz. The second data set reported in the table shows such correlations for annual returns on 97 randomly chosen US common stocks during the years 1948–1968. It is understood, of course, that mean–variance analysis is to be applied to the portfolio-as-a-whole rather than individual investments. Annual returns on individual stocks were used, however, as examples of return distributions with greater variability than that found in the portfolios reported in the prior column. As expected, the correlations were poorer for individual stocks than for mutual fund portfolios. For $U = \ln(1 + R)$, for example, the correlation was 0.880 for the annual returns on stocks as compared to 0.997 for the annual returns on the mutual funds.

Since monthly returns tend to be less variable than annual returns, we would expect the correlations to be higher for the former than the latter—and, indeed, they are. The correlations between EU and $f_{.01}$ for monthly returns on the same 97 stocks are shown in the fourth column in Table 2. For the logarithmic utility function, for example, the correlation is 0.995 for the monthly returns on individual stocks as compared to 0.880 for annual returns on the stocks and 0.997 for annual returns on the mutual funds. On the whole, the correlations for the *monthly* returns on individual stocks are comparable to those of the *annual* returns on the mutual funds.

As noted above, annual returns on individual stocks—i.e., on completely undiversified portfolios—have perceptibly smaller correlations between EU and $f_{.01}$ than do the annual returns on the well diversified portfolios of mutual funds. The final column in Table 2 presents correlations for “slightly diversified” portfolios consisting of a few stocks. Specifically, it shows the correlations $\rho_{.01}$ between EU and $f_{.01}$ on the annual returns for 19 portfolios of 5 or 6 stocks each randomly drawn (without replacement) from the 97 US stocks. We see that for the logarithmic utility function, $\rho_{.01} = 0.998$ for the random portfolios of 5 and 6, up from 0.880 for individual stocks. Generally, the $\rho_{.01}$ for the annual returns on the portfolios of 5 and 6 were comparable to those for the annual returns on the mutual funds. These results were among the most surprising of the entire analysis. They indicate that, as far as the applicability of mean–variance analysis is concerned, at least for joint distributions like the historical returns on stocks for the period analyzed, a little diversification goes a long way.

5. Highly risk-averse investors

The Levy–Markowitz results for the exponential with $b = 10$ differ markedly from those of the other utility functions reported in Table 2. In this section we explore the reason for this.

For $E = 0.1$ and $\sigma = 0.15$, Table 3 compares the exponential utility function with the quadratic approximation, Q_E in Eq. (2). The utility function is rescaled as follows

$$U = -1000e^{-10(1+R)}$$

With this scaling, the difference between $U(0.5)$ and $U(-0.3)$ is of the same order of magnitude as that for $\ln(1+R)$ in Table 1, namely, $0.41 - (-0.36) = 0.77$ in the latter case versus about 0.91 in the former. Table 3 is presented to four places, rather than two as in Table 1, since $U(R)$ rounds to 0.00 to two places for $R \geq 0.3$ for the exponential.

The first column of Table 3 lists R ; the second, $U(R)$; the third, the quadratic approximation Q_E ; the fourth column presents the difference between utility U and the quadratic Q_E , namely $d_E(R) = U(R) - Q_E(R)$. The table sheds light on why a quadratic approximation does much better for $\ln(1+R)$ than for $-\exp\{-10(1+R)\}$. In Table 1 the maximum difference between $\ln(1+R)$ and $Q_Z(R)$, is 0.02. In Table 3—with U scaled for comparability with Table 1—the absolute value of the difference, $|d_E|$, is 0.69 at $R = -0.3$. Thus the immediate reason that $f(E, V)$ is not a good approximation to EU in this case is because its Q_E is not a good approximation to $U(R)$. And the reason for that is that this $U(R)$ turns too quickly in the neighborhood of $R = E$. Between $R = -0.30$ and $R = 0.10$ utility increases by 0.90 from $U(-.3) = -0.912$ to $U(0.1) = -0.017$. But since $U \leq 0.0$ everywhere, it becomes comparatively flat as R increases further. Specifically, it rises less than 0.02 between $R = 0.10$ and “ $R = \infty$.” Essentially $U(R)$ has a knee at $R = E$.

Levy and Markowitz observe that an investor who had $-e^{-10(1+R)}$ as his or her utility function would have some strange preferences among probabilities of return. Since $U(R) < 0$ for all R , it follows that

$$0.5U(0.0) + 0.5U(R) < -0.0227 < U(0.1) \quad \text{for all } R.$$

Therefore, the investor would prefer

- (A) a 10% return with certainty, to
- (B) a 50–50 chance of zero return (no gain, no loss) versus a gain of 10⁹ percent or more.

Put another way, such an investor would prefer 10% with certainty to a 50–50 chance of either breaking-even or a “blank check.” Markowitz et al. (1994) find that real investors do not have such a low “value of a blank check,” (VBC). In their survey of brokerage customers, the median value of VBC was 401% as a fraction of the investor’s portfolio, or 143% as a fraction of the

investor’s total wealth, well above the less than 10% VBC of the exponential with $b = 10$.

6. Highly risk-averse investors and a risk-free asset.

Simaan (1993) explores the efficacy of MV-approximate maximization for investors with the exponential utility function when a risk-free asset is available versus when such a risk-free asset is not available. He finds that, for investors with exponential utility functions with large values of b , MV-approximate EU maximization is highly efficacious when a risk-free asset is available, and much less so when it is not.

In deriving these results, Simaan assumes that security returns follow a factor model,

$$1 + r_i = a_i + b_i F + u_i \quad i = 1, \dots, n \quad (17)$$

where the u_i are normally distributed, not necessarily independently, and F is a (skewed) random variable with a Pearson Type Three distribution. Simaan also assumes that the only constraint on portfolio choice is

$$\sum X_i = 1$$

Given these assumptions, Simaan is able to solve for the optimum portfolio.

Simaan illustrates these results based on monthly returns for ten randomly selected securities. The measure of efficacy used by Simaan is what he calls the “optimization premium,” namely, the return θ which would have to be added to the MV-approximate maximum portfolio in order to make it as desirable for the investor as the explicit optimum.

Table 4 presents the Simaan results. The first column shows the coefficient, b , in the exponential utility function; the second column shows the optimization premium when a risk-free asset is not available; the third column shows it when a risk-free asset is available. For example, for $b = 10$, if there is no risk-free asset one would have to add 0.00323, i.e., roughly 3/10 of 1% of the value of the portfolio each month to make it just as good as the explicitly maximized portfolio; whereas if a risk-free asset is available only 0.0001, i.e., one basis point per month need be added. Clearly, if one wishes to serve highly risk-averse investors, one should make a risk-free asset available.

7. Portfolios of call options

It is often said that mean–variance analysis is not valid if applied to portfolios which contain securities with asymmetric distributions, or whose returns vary in a nonlinear manner as a function of some security or index. Hlawitschka (1994) examines the efficacy of mean–variance and higher order approximations to expected utility for portfolios of calls, and concludes that the afore mentioned view is wrong. For his individual stock return

Table 3

Comparison of exponential utility with a quadratic approximation For $U = -1000e^{-10(1+R)}$; $E = 0.10$; and $\sigma = 0.15$.

R	$U(R)$	$Q_E(R)$	$U - Q_E$
−0.30	−0.9119	−0.2171	−0.6948
−0.20	−0.3355	−0.1420	−0.1935
−0.10	−0.1234	−0.0835	−0.0399
0.00	−0.0454	−0.0418	−0.0036
0.10	−0.0167	−0.0167	0.0000
0.20	−0.0061	−0.0084	0.0022
0.30	−0.0023	−0.0167	0.0144
0.40	−0.0008	−0.0418	0.0409
0.50	−0.0003	−0.0835	0.0832

Table 4

Simaan’s optimization premiums.

b	No risk-free asset	With risk-free asset
2	0.00023	0.00050
4	0.00073	0.00025
6	0.00144	0.00017
8	0.00229	0.00012
10	0.00323	0.00010
15	0.00581	0.00007
20	0.00859	0.00005
25	0.01147	0.00004
50	0.02646	0.00002
100	0.05719	0.00001

Table 5Spearman rank coefficients between $E[U(W)]$ and Taylor-series approximations.

Utility function	Portfolios of common stocks		Portfolios of calls	
	Second-order approximation	Third-order approximation	Second-order approximation	Third-order approximation
$U(W) = \log W$	1.0000	1.0000	0.9727	−0.8636
$U(W) = W^{0.1}$	1.0000	0.9909	0.9818	−0.8182
$U(W) = W^{0.5}$	1.0000	1.0000	0.9545	−0.2091
$U(W) = W^{0.9}$	1.0000	1.0000	0.9545	0.6909
$U(W) = -e^{-0.1W}$	1.0000	1.0000	0.9818	1.0000
$U(W) = -e^{-W}$	1.0000	1.0000	0.9273	−0.3545
$U(W) = -e^{-5W}$	0.9909	0.9909	0.8364	−0.8545
Average	0.9987	0.9974	0.9441	−0.2013

Table 6Correlations between $E[U(W)]$ and Taylor-series approximations.

Utility function	Second-order approximation	Third-order approximation	Fourth-order approximation	Fifth-order approximation	Sixth-order approximation
<i>Individual common stocks</i>					
$U(W) = \log(W)$	0.9986	0.9991	0.9990	0.9995	0.9997
$U(W) = W^{0.5}$	0.9997	0.9999	0.9999	0.9999	0.9999
$U(W) = -e^{-W}$	0.9989	0.9999	0.9999	1.0000	1.0000
<i>Individual call options</i>					
$U(W) = \log(W)$	0.8764	−0.7527	0.7304	−0.7054	0.7469
$U(W) = W^{0.5}$	0.7469	−0.5296	0.5628	−0.5503	0.5466
$U(W) = -e^{-W}$	0.7823	−0.6234	0.6193	−0.5858	0.5738

series, Hlawitschka uses the monthly returns for 10 years on 11 equally weighted portfolios each containing 20 randomly chosen stocks. His call return series assume that each call was 5% out-of-the-money when purchased and was priced according to the Black–Scholes model. Hlawitschka's call portfolios include a 10% position in T-Bills to avoid being wiped out in case all calls expire worthless.

Hlawitschka's principal results are shown in our Tables 5 and 6. The first column of Table 5 lists various utility functions; the second column shows the Spearman rank correlation coefficients between the actual average utilities and the quadratic Taylor-series approximations; the third column shows that between EU and a third order approximation. The final two columns of the table show such information for the portfolios of call options.

The results in Table 5 concerning portfolios of stocks seem even better than those in Levy–Markowitz. In particular, in six out of seven cases the correlation is perfect. In the seventh case, the exponential utility with $b = 5$, the rank correlation exceeds 0.99. One reason for these high correlations is that Hlawitschka uses monthly returns. As our Table 2 shows, MV approximations are quite close to EU for monthly return series, even for individual securities. Another reason for the high correlations is that these are rank correlations; therefore as long as EU and its MV-approximation produce the same ordering of the 11 portfolios, the rank order correlation is 1.0. A rationale for using the Spearman rank-order correlation coefficient, rather than the usual Pearson correlation coefficient as Levy and Markowitz did, is that if an approximation orders probability distributions correctly it will guide choice correctly.

The final two columns of Table 5 show that, as expected, the quadratic approximation to EU does less well for portfolios of call options than for portfolios of stocks. Nevertheless, the approximation is not bad. In the first five cases, the coefficient exceeds 0.95; in the sixth it exceeds 0.92; and in the troublesome exponential case with $b = 5$ the coefficient rounds to 0.84.

The real surprise in the table is the unreliability of the third order Taylor approximation. In particular, in five of the seven cases, the coefficient is negative! This is further explored by Hlawitschka in his Table 5, reproduced here as our Table 6, and his Table 6, (not reproduced here, but we will cite Hlawitschka's conclusions from it). In our Table 6, the first column lists utility functions; the

following five columns show rank correlations for second, third, fourth, fifth and sixth-order Taylor series approximations. The first three rows show such coefficients for returns on individual stocks (rather than portfolios); the second triplet of rows shows that for individual call options. For monthly returns on individual stocks, the quadratic approximation is quite good, with rank correlation coefficients exceeding 0.998 in all three cases, and with the higher order approximation doing even better.

The last three rows of Table 6 show that the quadratic approximation does less well than it did for individual stocks for monthly returns on portfolios consisting of 90% individual call options and 10% T-bills. Specifically, the three rank correlation coefficients round to 0.88, 0.75 and 0.78. But the 3rd and 5th order approximations were negatively correlated with EU, whereas the 4th and 6th order approximations had positive correlations, but ones inferior to the quadratic.

Hlawitschka's concluding paragraph is as follows:

Recall that, although a series may ultimately converge, little can be said about any of its partial series; convergence of a series does not imply that the terms immediately decrease in size or that any particular term is sufficiently small to be ignored. Indeed, it is possible, as demonstrated here, that a series may appear to diverge before ultimately converging in the limit. The quality of moment approximations to expected utility that are based upon the first few terms of a Taylor series, therefore, cannot be determined by the convergence properties of the infinite series. This is an empirical issue, and *empirically, two-moment approximations to the utility functions studied here perform well for the task of portfolio selection.* (Emphasis added).

8. Ederington's quadratic and Gaussian approximations to expected utility.

Ederington (1995) principal results are summarized in our Table 7, extracted from Ederington's Table 5. Ederington refers to the series represented in the table as “semi-ex ante simulated returns.” He argues that historical return series such as those used by Levy and Markowitz are an “ex post” sample drawn from some

Table 7
Correlations of Ederington's semi-ex ante simulated returns *MV* approximations.

Utility function	Normality assumption	Quadratic assumption	Actual four moments
$\text{Log}(b + R)$			
$b = 1.0$	0.9969	0.9993	0.9995
$b = 2.0$	0.9997	0.9999	0.9999
$b = 0.9$	0.9958	0.9984	0.9992
$(1 + R)^b$			
$b = 0.1$	0.9977	0.9995	0.9997
$b = 0.5$	0.9996	0.9999	0.9999
$b = 0.9$	0.9999	0.9999	0.9999
$(2 + R)^b$			
$b = 0.1$	0.9998	0.9999	0.9999
$b = 0.5$	0.9999	0.9999	0.9999
$-(1 + R)^{-b}$			
$b = 0.5$	0.9908	0.9961	0.9980
$b = 1.0$	0.9837	0.9864	0.9947
$b = 2.0$	0.9723	0.9289	0.9796
$b = 3.0$	0.9303	0.7893	0.9286
$b = 4.0$	0.7701	0.5325	0.7699
$b = 5.0$	0.4721	0.2324	0.4819
$-e^{-b(1+R)}$			
$b = 0.1$	0.9999	0.9999	0.9999
$b = 0.5$	0.9999	0.9999	0.9999
$b = 1.0$	0.9986	0.9995	0.9999
$b = 2.0$	0.9844	0.9979	0.9987
$b = 3.0$	0.9639	0.9907	0.9906
$b = 4.0$	0.9539	0.9664	0.9748
$b = 5.0$	0.9516	0.9201	0.9568
$b = 6.0$	0.9469	0.8526	0.9369

unknown population. Other possible samples from this same population might have randomly drawn more extreme returns than the one actual historical series, and therefore would have been less conducive to quadratic approximation. Ederington's solution is as follows:

[Q]uarterly return data for 130 mutual funds over the ten year period 1970–1979 were obtained from Wiesenberger Investment Companies Services' Management Results publication. For each fund i , we simulate 10,000 yearly returns by choosing four quarterly returns (r_1, r_2, r_3, r_4) at random from the set of 40 and calculating the yearly return R_i as:

$$(1 + R_i) = (1 + r_{i1})(1 + r_{i2})(1 + r_{i3})(1 + r_{i4})$$

This bootstrap like procedure may be termed a semi-ex ante (as opposed to purely ex post) method of comparing EU and MV. If market participants hold subjective probability distributions analogous to the distributions from which the quarterly returns are drawn and view these returns as serially independent, the set of simulated years should approximate their subjective distributions of yearly returns.

Ederington's synthetic series *do* include cases which are more challenging to MV-approximation than those of Levy and Markowitz's historical series. But, while details vary, our Table 7 shows that utility functions that did well in the Levy–Markowitz experiments also did well in Ederington's experiment.

The first column of Table 7 lists utility functions, including a number not included in the Levy–Markowitz study. The second and third columns show correlations with two different MV approximations. The second of these is based on the second-order Taylor series approximation in our Eq. (2); the first is a function of mean and variance based on what Ederington refers to as a “near normality assumption.” Specifically, Ederington's near-normality

assumption uses the fourth-order Taylor approximation, but rather than using the third and fourth moments

$$M_3 = E(R - E)^3$$

$$M_4 = E(R - E)^4$$

of the actual return distribution, he assumes

$$M_3 = 0$$

$$M_4 = 3V$$

as is the case for a normal (Gaussian) distribution. This provides a “fourth-order approximation” which is a function of E and V only.

Kroll et al. (1984) show that many of the return series they used are probably not drawn from normal distributions. Ederington's comment on such results is essentially this: true, return distributions are not exactly normal; but neither are utility functions exactly quadratic. Therefore leave it up to the data to tell us which method of approximation works best.

The final column of Table 7 shows correlations of EU with the expected value of a fourth-order Taylor approximation using the distribution's actual M_3 and M_4 . The table reports the correlations for Ederington's 22 utility functions to four places, as he did, for the reader's detailed examination. My own “take away” from Ederington's table is as follows: in 14 of the 22 cases, the MV approximation based on the quadratic in our Eq. (2) had a correlation with EU over 0.99. In 12 of these 14 cases, the approximation based on near-normality also had a correlation exceeding 0.99, including four cases in which all three approximations had a correlation of 0.9999. In the other two cases (in which the quadratic approximation had a correlation exceeding 0.99) the near-normality approximation had correlations rounding to 0.98 and 0.96. Thus, in the 14 cases in which MV quadratic approximation did exceedingly well, the near-normality approximation did about as well, and there was little room for improvement by using the distribution's actual M_3 and M_4 .

This leaves eight cases in which the MV quadratic approximation had a correlation with EU of less than 0.99. Ignoring the case in which both the quadratic and near-normality approximations have correlations in excess of 0.98, we are left with seven “challenging” cases. An interesting pattern appears in each of these. While all three approximations deteriorate:

- The quadratic approximation does worse than the other two, and
- The near-normality approximation does roughly as well as the fourth-degree approximation using the series' actual M_3 and M_4 .

For example, for $-\exp\{-6(1 + R)\}$ the correlations (for the nearly normal, quadratic and fourth-order approximations respectively) are

$$0.95 \quad 0.85 \quad 0.94$$

whereas for $-(1 + R)^{-5}$ they are

$$0.47 \quad 0.23 \quad 0.48.$$

Thus as b increases, the near-normality approximation holds up better than the quadratic approximation, and does roughly as well as the fourth degree approximation using the actual M_3 and M_4 . Therefore, confining ourselves to Ederington's three approximations to expected utility, there is little advantage to considering moments higher than mean and variance. For utility functions with medium to high VBC the quadratic approximation does so well that there is little room for a fourth-order approximation to do perceptibly better; whereas for utility functions with low VBC, the

near-normality approximation does about as well as the fourth-order Taylor approximation using the distribution's own M_3 and M_4 .

9. Mean–variance approximations to the geometric mean

Markowitz (2012a) presents historical comparisons between the geometric mean G and six mean–variance approximations to it, for two data sets: one consisting of the historical returns on asset classes widely used in asset allocation decisions, the other containing the real returns during the 20th Century of the equity markets of sixteen countries. I will refer to the first of these as the AC (Asset Class) database, and the second as the DMS (Dimson et al., 2002) database. Both were obtained through Morningstar's EnCorr databases and software. The six mean–variance approximation methods are labeled QO , QE , EDD , LMT , NLN and HL . The approximate values computed using these six methods will be labeled similarly as g_{QO} , g_{QE} , etc.

QO and QE are obtained by exponentiating the two approximations to $ELn(1 + R)$ presented in our Eqs. (5) and (6), namely,

$$g_{QO} = \exp\left\{E - \frac{1}{2}(\sigma^2 + E^2)\right\} - 1 \quad (18a)$$

$$g_{QE} = \exp\left\{Ln(1 + E) - \frac{1}{2}\sigma^2/(1 + E)^2\right\} - 1 \quad (18b)$$

(In calculations with historical returns such as reported here, E represents historical average rather than expected value.) The EDD approximation is

$$g_{EDD} = \exp\left\{Ln(1 + E) - \frac{1}{2}\sigma^2/(1 + E)^2 - .75\sigma^4/(1 + E)^4\right\} - 1 \quad (18c)$$

This is the exponentiated version of the “fourth-degree” MV approximation to $ELn(1 + R)$ proposed by Ederington as described in the preceding section. The next approximation is

$$g_{LMT} = E - \frac{1}{2}V \quad (18d)$$

This simple formula—often used for “back of the envelope” calculations—follows from Eq. (18a) if one assumes that E^2 and G^2 are negligible. I refer to it as the LMT (for limit) approximation. The next approximation equals what G would be if $Ln(1 + R)$ were normally distributed, namely,

$$1 + g_{NLN} = (1 + E)/\sqrt{1 + V/(1 + E)^2} \quad (18e)$$

I refer to this as the NLN (near log-normal) approximation. The final approximation to G as a function of E and V is what G would be for a distribution with two equally likely outcomes, namely,

$$(1 + g_{HL})^2 = (1 + E)^2 \pm \sigma \quad (18f)$$

Jean and Helms (1983) ascribes this to Henry Latané (1959).⁴ I therefore label it the HL approximation.

Despite the encouraging approximation reported in Markowitz (1959) as shown in our Table 1, Markowitz (2012a) eliminates g_{QO} early as clearly the worst of the lot; and then the g_{LMT} and finally g_{EDD} are eliminated after further examination. No further pruning seemed justifiable, since g_{HL} did a bit better than the remaining other two for the AC database, whereas g_{NLN} did so for the DMS database, and Markowitz (2012a) shows that necessarily

$$g_{NLN} \geq g_{QE} \geq g_{HL}$$

⁴ Jean and Helms cite Latané (1959). While Latané discusses the two-outcome case, I do not find Eq. (10f) in this reference. I do find it in Latané and Tuttle (1967) with a different justification.

10. Other measures of risk

As described in the preceding section, Markowitz (2012a) compared six different MV methods for approximation $ELn(1 + R)$. Among these the Markowitz (1959) approximation in Eq. (6) here did as well as any. Markowitz (forthcoming) asks could another of the commonly proposed risk-measures have done better. The measures compared in Markowitz (forthcoming) are:

Variance (V),
Mean Absolute Deviation (MAD),
Semivariance (SV),
Value at Risk (VaR),
Conditional Value at Risk ($CVaR$).

From Eq. (6) we see that

$$Ln(1 + E) - Ln(1 + g_{QE}) = \frac{1}{2}V/(1 + E)^2 \quad (20)$$

Thus the expression on the right hand side of Eq. (20) characterizes the f_E method for approximating the difference on the left. Let

$$\Delta L \equiv Ln(1 + E) - Ln(1 + G) \quad (21)$$

Markowitz (forthcoming) tests approximations to ΔL of the form

$$\Delta L = \beta \cdot f(RM) \quad (22)$$

where $f(RM)$ represents a function of some risk-measure. (Here $f(RM)$ corresponds to the $f(E, V)$ notation earlier, but with E now implicit.) For each of the two databases in turn, the beta in Eq. (22) was fit by regressing the ΔL s of the various series against the corresponding values of $f(RM)$. The regressions have zero intercept, since $E = G$, therefore $\Delta L = 0$, if a series has no variability.

The $f(RM)$ considered by Markowitz (forthcoming) are listed in the first column of Table 8. RawVaR is computed as if the returns in each data series were equally likely and were the only possible returns in the population. Thus RawVaR at the 5% level is the largest loss such that this loss—plus all returns which are worse than it—constitute at least 5% of the population. For a small data series, such as the Emerging Market series in the AC database, there may be a considerable gap between RawVaR and the next lower return. Interpolated VaR assumes, instead, that the return distribution has a step-function probability-density with returns uniformly distributed between -RawVar and the next lower return.

Interpolated VaR was not computed for the DMS database. Since each series in the DMS database has 101 observations, the fifth

Table 8
Root mean-squared errors (percent).

$f(RM)$	Frequently used asset classes	Real returns for 16 countries 1900–2000
Variance	0.10	0.17
Variance/ $(1 + E)^2$	0.05	0.18
MAD	0.51	0.70
MAD ²	0.29	0.40
MAD ² / $(1 + E)^2$	0.20	0.42
Semi variance	0.11	0.35
Semivariance/ $(1 + 2)^2$	0.12	0.35
RawVaR	0.65	0.68
$((\text{Raw VaR} + E)/K)^2$	0.38	0.61
$((RV + K)/K)^2/(1 + E)^2$	0.46	0.60
InterpVaR	0.32	–
$((\text{Int VaR} + E)/K)^2$	0.38	–
$((IV + E)/K)^2/(1 + E)^2$	0.30	–
CVaR	0.48	0.55
CVaR ²	0.15	0.49
CVaR ² $1/(1 + E)^2$	0.17	0.46

from the worst return was used to define VaR. This makes VaR be at the 5/1.01% level, and RawVaR precisely equal to VaR. For both databases CVaR was computed as the average return given that return equaled -RawVaR or worse. It was deemed unnecessary to compute CVaR using both RawVaR and interpolated VaR where these differed, since there is a large overlap in the range of the two computations.

The second and third columns of Table 8 show the root-mean-squared (RMSQ) error made by each tested $f(RM)$ for each of the two databases. The second column of the table lists the RMSQs in the AC database, and the third shows the same for the DMS database. RMSQ is expressed as a percent. For example, using a confidence interval equal to the estimate plus or minus two RMSQ, in the AC database if “adjusted variance”—i.e., $V/(1+E)^2$ —estimated a geometric mean of 10.0%, this estimate would be subject to an error of probably no more than $\pm 2 \cdot (0.05) = 0.1\%$, i.e., 10 bps (where a basis point, bp, is 1/100th of 1%). MAD, on the other hand, has an RMSQ of 0.51, therefore is subject to an estimated error of probably no more than plus or minus 102 bps, slightly over 1% point. Adjusted MAD-squared does much better, with an RMSQ 0.20, therefore a confidence interval of ± 40 bps, about twice that of variance and four times that of adjusted variance.

Viewing the entire second column we see that the best fit in the AC database is provided by adjusted variance with an RMSQ of 0.05, followed by (unadjusted) variance, semivariance, adjusted semivariance and CVaR² with RMSQs of 0.10, 0.11, 0.12 and 0.15 respectively.

All $f(RM)$ did worse in the DMS database than they did in the AC database. In particular, in this database unadjusted variance does slightly better than adjusted variance, with RMSQs equal to 0.17 and 0.18. The next closest risk measure is semivariance, adjusted and not, with RMSQs of 0.35, about twice that of variance. MAD-squared and adjusted MAD-squared are a bit behind the semivariance measures. The worst performers are unadjusted MAD and the various functions of VaR and CVaR with RMSQs ranging from 0.46 to 0.70.

Thus, in the DMS database with its large losses, functions of VaR and CVaR—which are promoted as the measures-to-use in case of large deviations—have substantially larger errors of approximation than do functions of variance.

11. Other pioneers

In previous sections I have tried to review what I consider to be the most critical steps towards our present understanding of MV approximations of EU, these include: the Levy, Markowitz observation that MV approximations to EU are quite good for many $U(R)$ —except those with extremely high risk-aversion, i.e., implausibly low VBC; Simaan’s observation that such investors will nevertheless be well served if the MV analysis includes a risk-free asset; Hlawitschka’s observation that MV approximations work for portfolios of calls; Ederington’s observation that functions of mean and variance other than ones based on a quadratic approximation to $U(R)$ may work well for utility functions with implausibly low VBC, even without a risk-free asset; and my own recent observations on MV’s better job with large deviations than approximations using semivariance, MAD, VaR and CVaR. Along the way I have skipped over important contributors because, for example, their contributions principally confirmed results already obtained with other databases or points-of-view. This section attempts to fill in some of these omissions.

Young and Trent (1969) were the first, after Markowitz (1959), to explore the efficacy of various mean–variance (and higher order) approximations to $ELn(1+R)$ or, equivalently, to the geometric mean G . Based on monthly historical return series, they concluded

that the Markowitz (1959) quadratic approximation in our Eq. (2) did as well as any of the other quadratic approximations they tried, and higher order approximations added little.

Levy and Markowitz (1979), discussed in a previous section, assumed that if EU and $f(E,V)$ were highly correlated, then $f(E,V)$ would provide near maximum EU. Kroll et al. (1984) used the historical returns on 20 randomly selected stocks to explore how close maximum $f(E,V)$ is to explicitly maximized EU among portfolios on an MV-efficient frontier. They found that it does quite well.

How to measure efficacy of approximation has been an ongoing topic in this literature. In particular, Dexter et al. (1980) use the percent difference in “certainty-equivalent” wealth as their measure of MV efficacy. The Dexter et al. conclusion was that “In all cases $[W_C^E \text{ and } W_C^M]$ were virtually identical,” i.e., the maximum certainty equivalent was little greater than that of the mean–variance approximation.

Hakansson (1971) explored the efficacy of mean–variance approximation to $ELn(1+R)$ for portfolios made up of one risk-free asset and two hypothetical stocks whose returns—one with probabilities of 100% loss vs. 50% gain, the other with probabilities of 15% gain vs. 165% gain—are well beyond the range where the quadratic in Table 1 fits well. Not surprisingly, Hakansson finds that mean–variance approximations are not efficacious in this case. Generally, analyses based on historical portfolio returns reported that MV approximations did well. Those which used made-up return distributions (of which Loistl’s is the most extreme) found distributions where MV approximations broke down.

Grauer (1986) considers the efficacy of MV approximation to the power or logarithmic utility function when the investor can borrow all he or she wants at the risk-free rate, as permitted in the Sharpe–Lintner Capital Asset Pricing Model (see Mossin, 1966). Grauer reports unfavorable results unless constraints are added to the mean–variance optimization to avoid bankruptcy. The Simaan analysis, reported above, also found poor results for the exponential utility function for b smaller than the $b \geq 2$ cases reported in Table 6. He observed that the portfolios in these cases involved, simultaneously, extremely large short and long positions. He concluded that the constraint set he used—without nonnegativity constraints to permit analytic solution—did not produce real-world-relevant results for these values of b .

Pulley (1983) found that

investors maximizing expected logarithmic utility would hold virtually the same portfolios as investors maximizing certain mean–variance function. . . [based on] exact empirical distributions of security returns. Moreover, we find the goodness of the mean–variance approximations to be robust for different holding periods, different ratios of noninvested to invested wealth, and different subjective return distributions, including nonnormal distributions.

12. Conclusion

It is now over a half-century since Markowitz (1959) first defended MV analysis as a practical way to approximately maximize EU. In light of repeated confirmation since then of the efficacy of MV approximations to EU, the persistence of the Great Confusion—that MV analysis is applicable in practice only when return distributions are Gaussian or utility functions quadratic—is as if geography textbooks of 1550 still described the Earth as flat.

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